1st Place: Aspect 4 – Make mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them

- **Analysis**: This is the most central operation in this problem. The shape of the oil spill on the map is irregular. To estimate its area, students need to actively "construct" a geometric approximation that is easier to calculate, or systematically extract area information from the original image through some method.
- **Manifestation in the Question**: Typical strategies include: 1) **Overlay/Approximation Method**: Visually or with a pencil, approximate the irregular shape on the map as a regular shape (e.g., an ellipse, a combination of rectangles/triangles). 2) **Grid-Counting Method**: Overlay (or imagine overlaying) a transparent grid on the map and count the number of squares (full and partial) covered by the spill. Both involve "making" a new, measurable mathematical representation and extracting key data from it (dimensions of the approximate shape or grid square counts).

2nd Place: Aspect 7 – Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- **Analysis**: Students need to directly manipulate graphical information (the shape on the map, the scale) and numerical information (scale value, calculation results).
- **Manifestation in the Question**: The operational flow includes: identifying the map scale and understanding its meaning; measuring or estimating the dimensions of the constructed approximate shape (e.g., the lengths of the major and minor axes of an approximate ellipse on the map); performing unit conversion (converting lengths in cm/mm on the map to actual km lengths using the scale); and finally performing area calculations. The entire process tightly intertwines graphical manipulation and numerical operations.

3rd Place: Aspect 12 – Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- **Analysis**: Getting from reading the map to a final area estimate involves a clear multi-step procedure.
- **Manifestation in the Question**: A reasonable sequence of steps is:
  1. **Step A (Modeling)**: Choose an estimation strategy (e.g., approximate as an ellipse).
  2. **Step B (Measurement)**: Measure the key dimensions of this approximate shape on the map (e.g., length of major axis `a`, minor axis `b` in cm).
  3. **Step C (Conversion)**: Use the map scale (e.g., 1 cm : 10 km) to convert map dimensions to real-world dimensions (A = a * scale factor [km/cm], B = b * scale factor [km/cm]).
  4. **Step D (Calculation)**: Apply the appropriate area formula (e.g., ellipse area = π * A ∗ B / 4).
  5. **Step E (Presentation)**: State the final estimated area in km^2.

4th Place: Aspect 6 – Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

- **Analysis**: Solving the problem relies on the accurate application of several core mathematical concepts and rules.
- **Manifestation in the Question**: Key rule systems include:
  - **Concept and conversion rule of scale**: Understanding the linear proportional relationship between lengths on the map and actual lengths.
  - **Area formulas**: Using the correct area formula based on the chosen approximation (e.g., rectangle, circle, ellipse, or "each square represents xx km^2" when using grid counting).
  - **Unit conversion**: Ensuring consistency and correct conversion of length and area units.

5th Place: Aspect 13 – Use an understanding of the context to guide or expedite the mathematical solving process

- **Analysis**: The word "estimate" and the real-world context of an "oil spill" strongly imply the nature and precision requirements of the solution.
- **Manifestation in the Question**: The context informs students that this is an **estimation problem**, not an exact calculation. Therefore, they can employ reasonable approximation methods (e.g., treating the shape as an ellipse), and the result can be a range or approximate value, not needing absolute precision. This guides their tendency in strategy selection and their attitude towards the result.

6th Place: Aspect 10 – Make sense of, relate and use a variety of representations when interacting with a problem

- **Analysis**: The problem contains **visual-graphical representations** (map with scale, irregular oil spill shape) and **symbolic-numerical representations** (scale numbers, length measurements, area formulas, final answer).
- **Manifestation in the Question**: Students must establish connections between them: translating the visual form on the graphic into numerical dimensions usable for calculation, and applying the abstract mathematical relationship of the scale to the specific measured values.

7th Place: Aspect 8 – Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- **Analysis**: A complete solution process should demonstrate key intermediate steps to make the reasoning clear.
- **Manifestation in the Question**: In their solution, students should show: the approximation method used (e.g., "I approximated the spill as an ellipse"), the measurement data obtained from the map, the process of applying the scale for conversion, and the calculation steps substituting into the formula. This demonstrates the ability to organize and communicate the solution process.

8th Place: Aspect 11 – Switch between different representations in the process of finding solutions

- **Analysis**: The problem-solving process is essentially a series of conversions between different representations.
- **Manifestation in the Question**: Converting from a **graphical representation** (oil spill outline) to a **simplified graphical model** (e.g., an ellipse), then to a **numerical representation** (measured length, width); using the **scale rule** (another representation) to convert map numerical values to real-world values; finally applying a **formula system** (symbolic representation) to arrive at the final numerical answer.

第1位：方面 4 - 制作数学图表、图形或构造物，并从中提取数学信息

- **分析**：这是本题最核心的操作。地图上的油污形状是不规则的。为了估算面积，学生需要主动“构造”一个便于计算的几何近似图形，或通过某种方法从原图中系统地提取面积信息。
- **题目体现**：典型的策略包括：1) **覆盖法**：用眼睛或铅笔在图上将不规则形状近似为规则图形（如椭圆、多个矩形、三角形组合）；2) **网格计数法**：在图上（或想象中）覆盖透明方格纸，数出油污覆盖的方格数（完整和部分）。这都是在“制作”一个新的、可测量的数学表示，并从中提取关键数据（近似图形的尺寸或方格数）。

第2位：方面 7 - 处理数字、图形与几何表示

- **分析**：学生需要直接处理图形信息（地图上的形状、比例尺）和数字信息（比例尺数值、计算结果）。
- **题目体现**：操作流程包括：识别地图比例尺并理解其含义；测量或估算所构造的近似图形的尺寸（如近似的椭圆长轴和短轴在地图上的长度）；进行单位换算（将地图上的厘米/毫米长度通过比例尺转换为实际公里长度）；最后进行面积计算。整个过程紧密交织着图形处理和数字运算。

第3位：方面 12 - 使用多步骤流程得出数学解答或结论

- **分析**：从读图到得出最终的面积估算值，涉及一个清晰的多步流程。
- **题目体现**：一个合理的步骤序列是：
  1. **步骤A（建模）**：选择估算策略（如近似为椭圆）。
  2. **步骤B（测量）**：在地图上测量该近似图形的关键尺寸（如长轴a、短轴b的长度，单位：cm）。
  3. **步骤C（换算）**：利用比例尺（例如，1 cm : 10 km）将地图尺寸换算为实际尺寸（A = a * 比例系数 [km/cm]， B = b * 比例系数 [km/cm]）。
  4. **步骤D（计算）**：应用相应图形的面积公式（如椭圆面积 = π * A * B / 4）。
  5. **步骤E（表述）**：给出最终估算面积（km^2）。

第4位：方面 6 - 理解并运用基于定义、规则和形式系统的构造

- **分析**：解决问题依赖于对几个核心数学概念和规则的准确运用。
- **题目体现**：关键的规则系统包括：
  - **比例尺的概念与换算规则**：理解地图上长度与实际长度的线性比例关系。
  - **面积公式**：根据所选近似图形，运用正确的面积公式（如矩形、圆形、椭圆、或通过方格计数法时的“每个方格代表xx km^2”）。
  - **单位换算**：确保长度和面积单位的一致性和正确换算。

第5位：方面 13 - 利用对情境的理解来指导或加速求解过程

- **分析**：“估算”一词和“油污扩散”这一现实情境强烈暗示了解题的性质和精度要求。
- **题目体现**：情境告诉学生这是一个**估算问题**，而非精确计算。因此，他们可以采用合理的近似方法（如将形状视为椭圆），并且结果可能是一个范围或近似值，不需要绝对精确。这指导了他们选择策略的倾向和对待结果的态度。

第6位：方面 10 - 在解决问题时理解、关联和使用多种表征

- **分析**：题目包含了**视觉-图形表征**（带比例尺的地图、不规则油污形状）和**符号-数字表征**（比例尺数字、长度测量值、面积公式、最终答案）。
- **题目体现**：学生必须在两者间建立联系：将图形上的视觉形态转化为可用于计算的数字尺寸，并将比例尺这个抽象的数学关系应用到具体的测量值上。

第7位：方面 8 - 阐明解决方案，展示和/或总结、呈现中间数学结果

- **分析**：一个完整的解答过程应当能够展示关键的中间步骤，以使推理清晰。
- **题目体现**：在解答中，学生应展示：所使用的近似方法（如“我将油污近似为一个椭圆”）、从图上获得的测量数据、应用比例尺进行换算的过程、代入公式的计算步骤。这体现了对解决过程的组织和表达能力。

第8位：方面 11 - 在求解过程中在不同表征间转换

- **分析**：解题过程本质上是在不同表征间的一系列转换。
- **题目体现**：从**图形表征**（油污轮廓）转换为**简化图形模型**（如椭圆），再转换为**数值表征**（长、宽测量值）；利用**比例尺规则**（另一种表征）将地图数值转换为实际数值；最后应用**公式系统**（符号表征）得出最终的数字答案。